

Trader Performance Analysis Using Bitcoin Market Sentiment

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1. Introduction

Financial markets are heavily influenced by investor sentiment, especially in cryptocurrency trading where emotions often drive price movements. The Bitcoin Fear & Greed Index measures market psychology by classifying daily sentiment into categories such as **Fear** and **Greed**.

The objective of this project is to analyze whether Bitcoin market sentiment affects trader performance on the Hyperliquid trading platform. By combining historical trading data with the Fear & Greed Index, the analysis aims to identify patterns in profitability, trading behavior, and risk-taking that can support more informed trading decisions.

2. Methodology

The analysis was performed using Python with the Pandas, NumPy, Matplotlib, and Seaborn libraries.

The workflow consisted of the following steps:

- Loaded the Historical Trader dataset and the Bitcoin Fear & Greed dataset.
- Cleaned the datasets by handling missing values, removing duplicate records, and converting timestamp fields into a common date format.
- Merged both datasets using the trading date as the common key.
- Conducted Exploratory Data Analysis (EDA) to examine trading behavior across different market sentiment conditions.
- Created visualizations including:

Market sentiment distribution

Closed PnL distribution

Average profit by sentiment

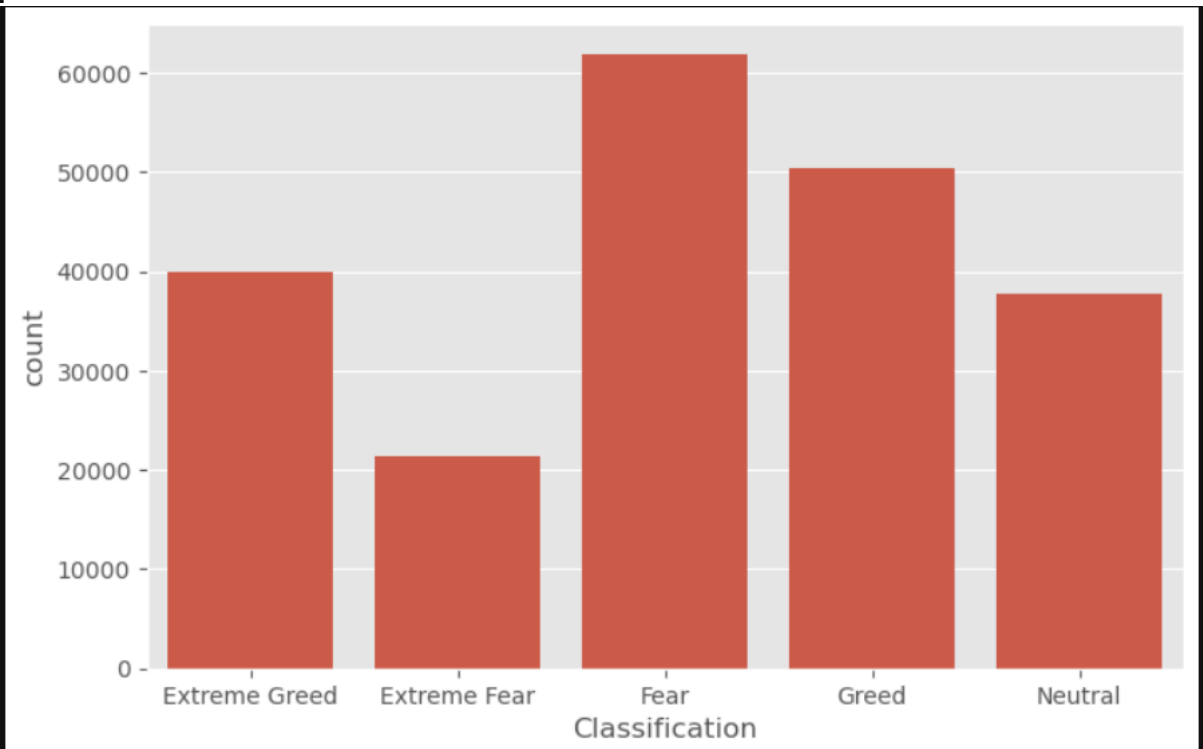
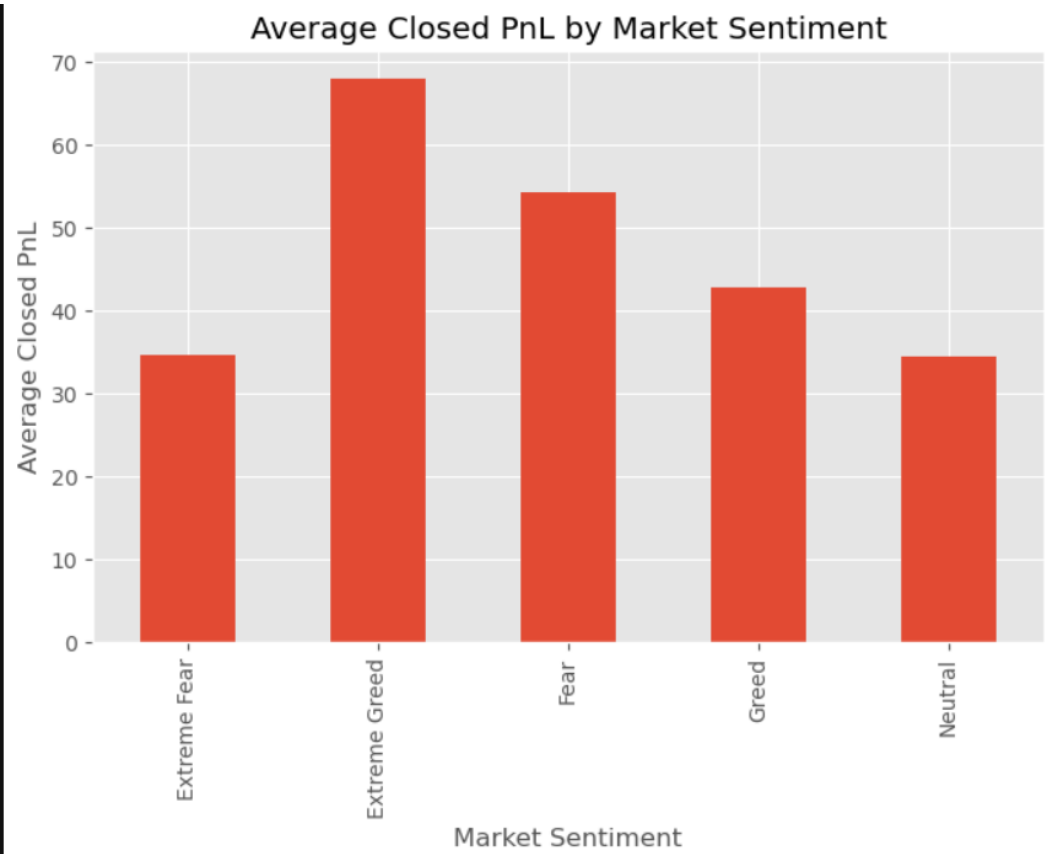
Buy vs. Sell distribution

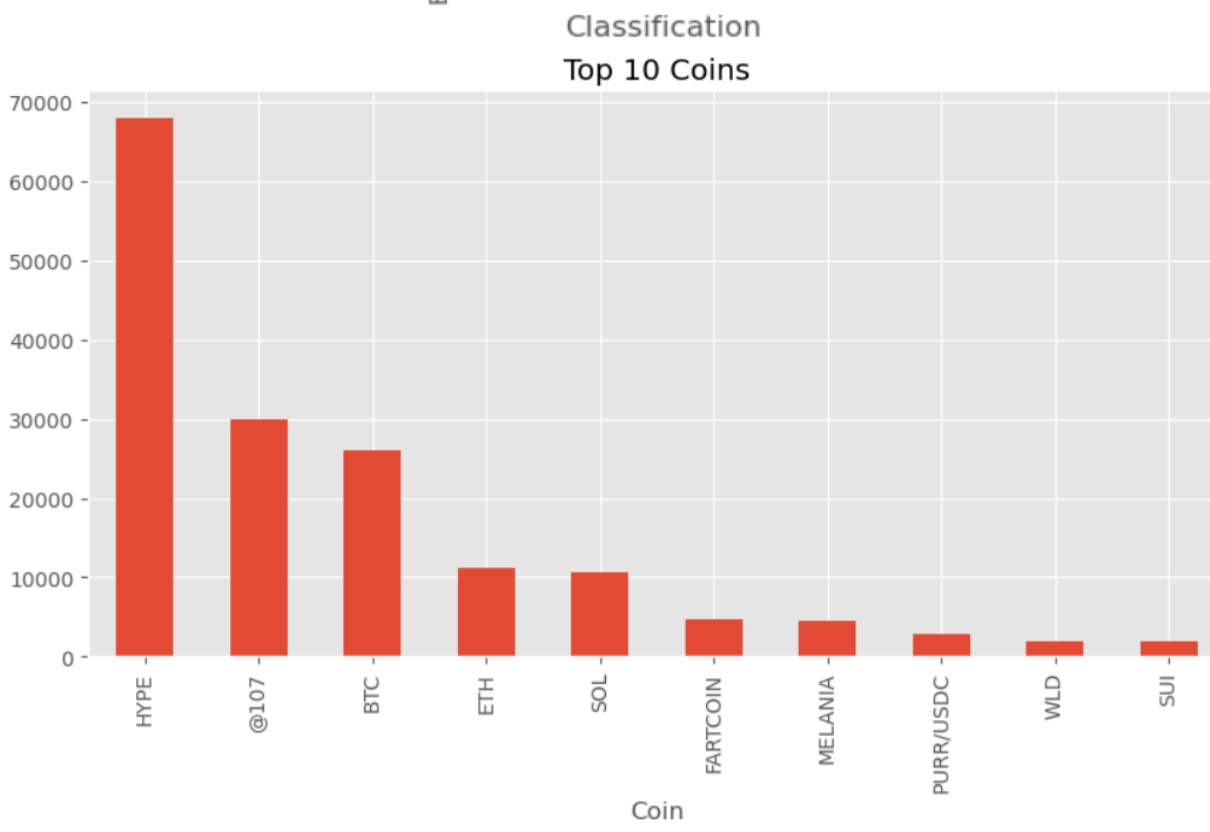
Top traded coins

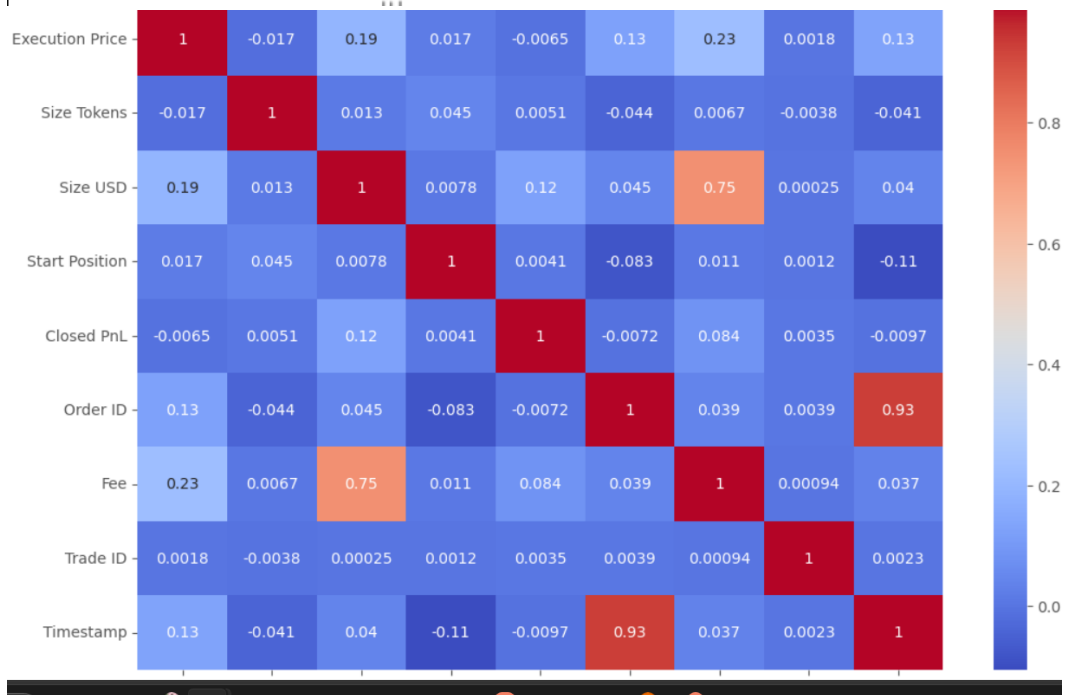
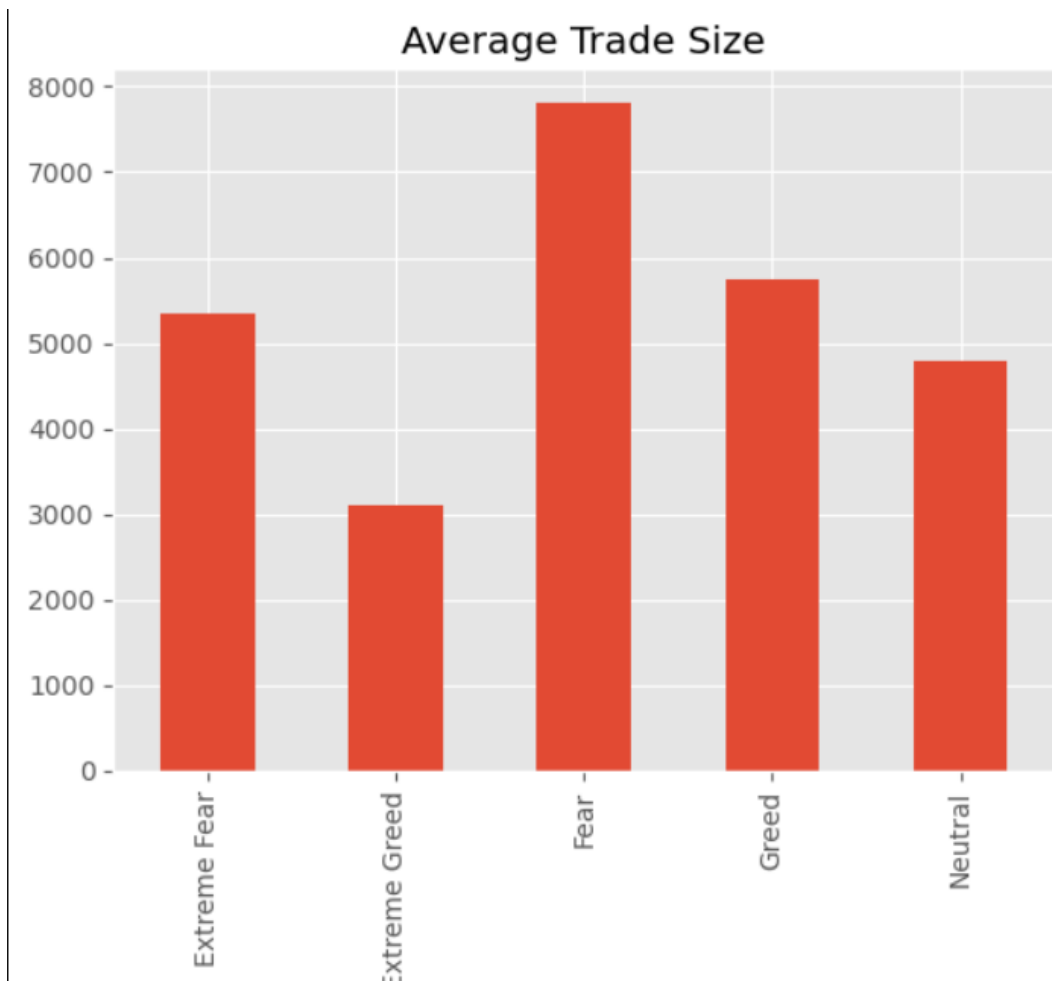
Top profitable traders

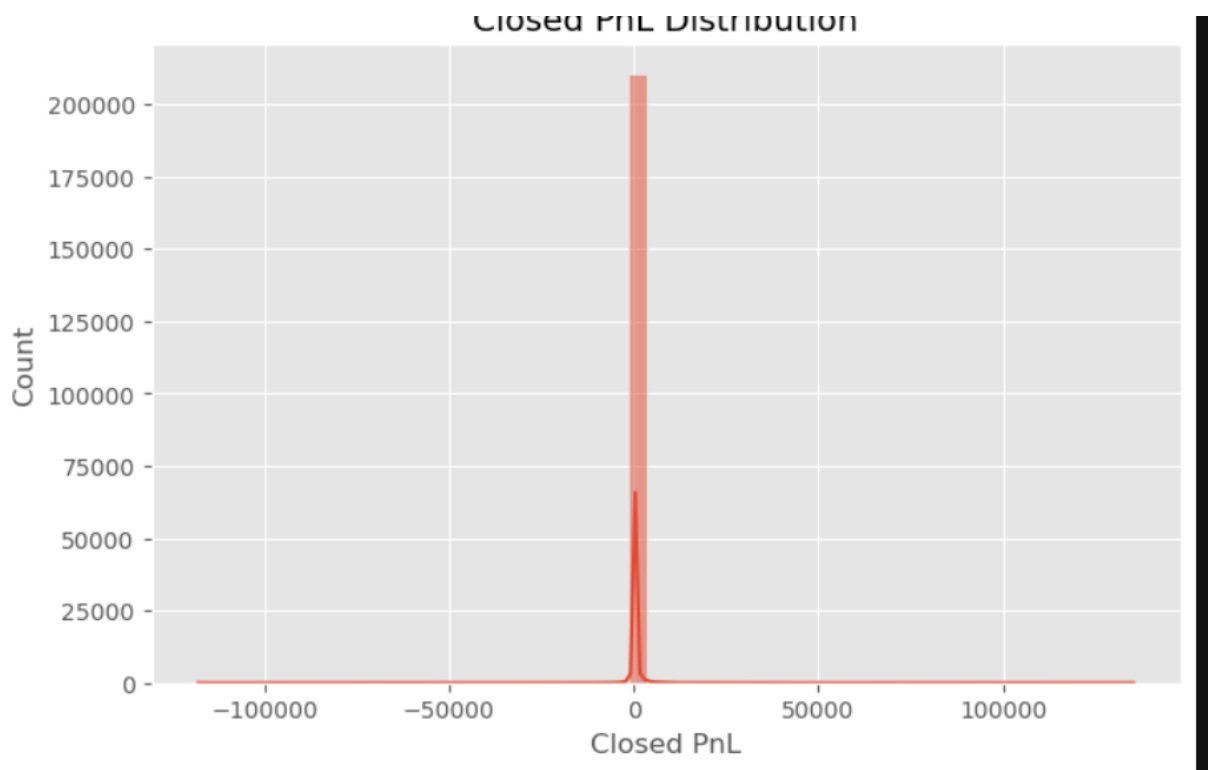
Correlation heatmap

Trade size analysis









3. Results

The merged dataset enabled comparison of trader performance under different market sentiment conditions.

Key observations include:

- The trading dataset contained over **211,000 trade records** from multiple trading accounts.
- Market sentiment varied between Fear and Greed over the observation period.
- Average Closed PnL differed across sentiment categories.
- Trading activity was concentrated in a small number of highly traded cryptocurrencies.
- A limited number of traders generated a significant portion of the total profits. Correlation analysis highlighted relationships between leverage, trade size, and profitability.

4. Key Insights

The analysis revealed several important findings:

- Market sentiment influences trading behavior and can affect overall profitability.
- Traders often modify position sizes and leverage according to prevailing market sentiment.
- Higher leverage may lead to greater profit potential but also increases downside risk.
- Certain cryptocurrencies consistently account for the majority of trading activity.
- Profit distribution is highly uneven, with a relatively small group of traders contributing most of the cumulative gains.
- Incorporating sentiment indicators alongside technical analysis may improve decision-making and risk management.

5. Conclusion

This project demonstrates that combining market sentiment data with historical trading records provides valuable insights into trader performance. The Bitcoin Fear & Greed Index can serve as an additional decision-support tool by helping traders recognize shifts in market psychology.

Future work could include developing predictive machine learning models to estimate trade profitability based on sentiment, leverage, trade size, and other market features. Integrating these models into automated trading systems may further improve strategy performance and risk control.